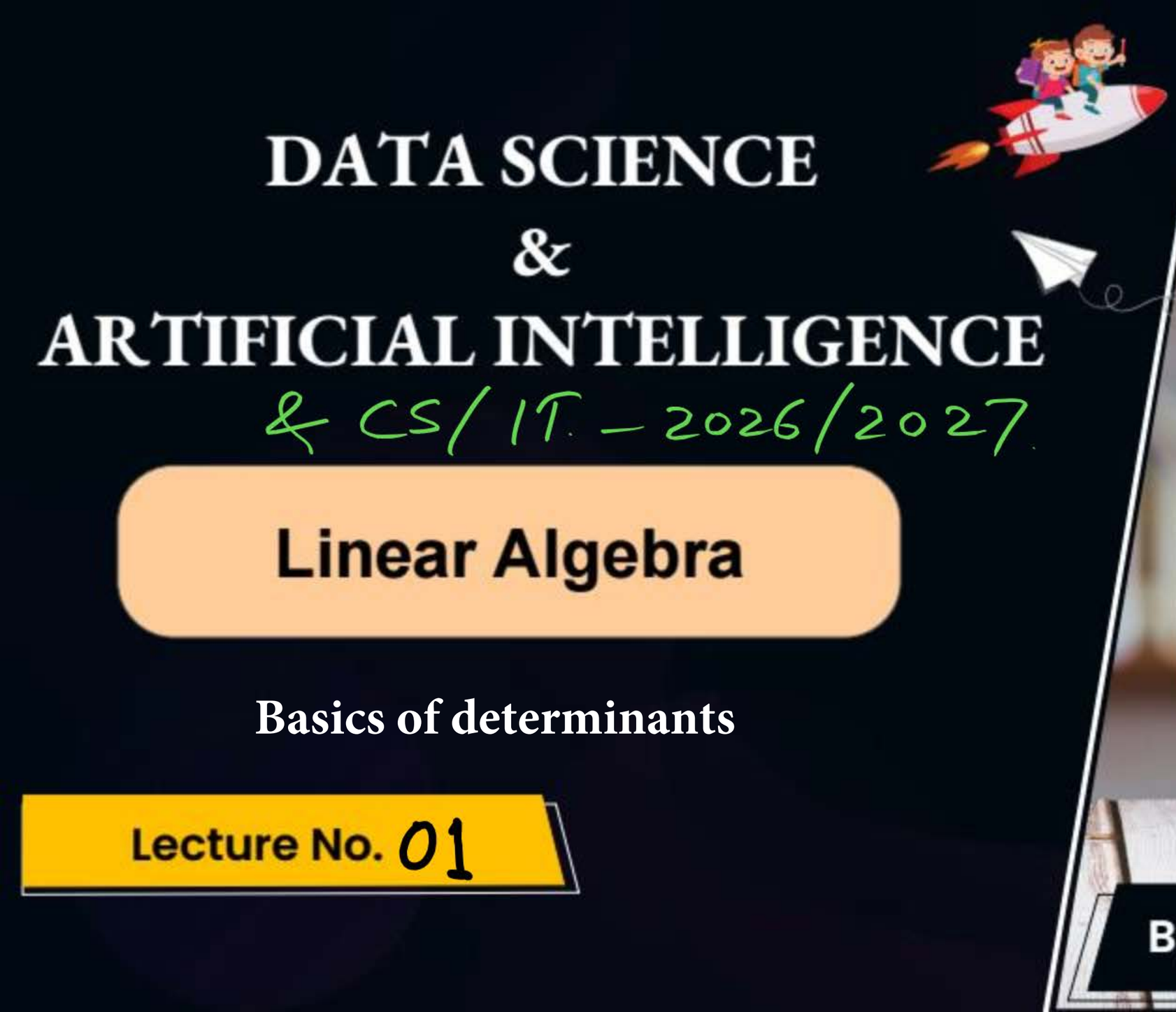
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT – 2026/2027

Linear Algebra

Basics of determinants

Lecture No. 01



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

INTEGRATION

Topics to be Covered



Topic

1. Application of Integration in finding
   - (i) Length of Curve
   - (ii) Volume of Solid formed by Revolution
2. Determinants (BASIC)

Error: Let In the Calculation of Volume error $= dV$ then.

$\because V \longrightarrow (x, y, z)$

$$\Rightarrow \boxed{dV = \left(\frac{\partial V}{\partial x}\right)dx + \left(\frac{\partial V}{\partial y}\right)dy + \left(\frac{\partial V}{\partial z}\right)dz}$$

Absolute Error $= dV$

Relative " $= \frac{dV}{V}$

% " $= \left(\frac{dV}{V} \times 100\right)\%$

Truncation Error: eg $e^{0.2} = 1 + (0.2) + \frac{(0.2)^2}{2!} + \frac{(0.2)^3}{3!} + \cdots$

$\cong 1 + (0.2) + \frac{(0.2)^2}{2!} +$ Neglect

T. Error.

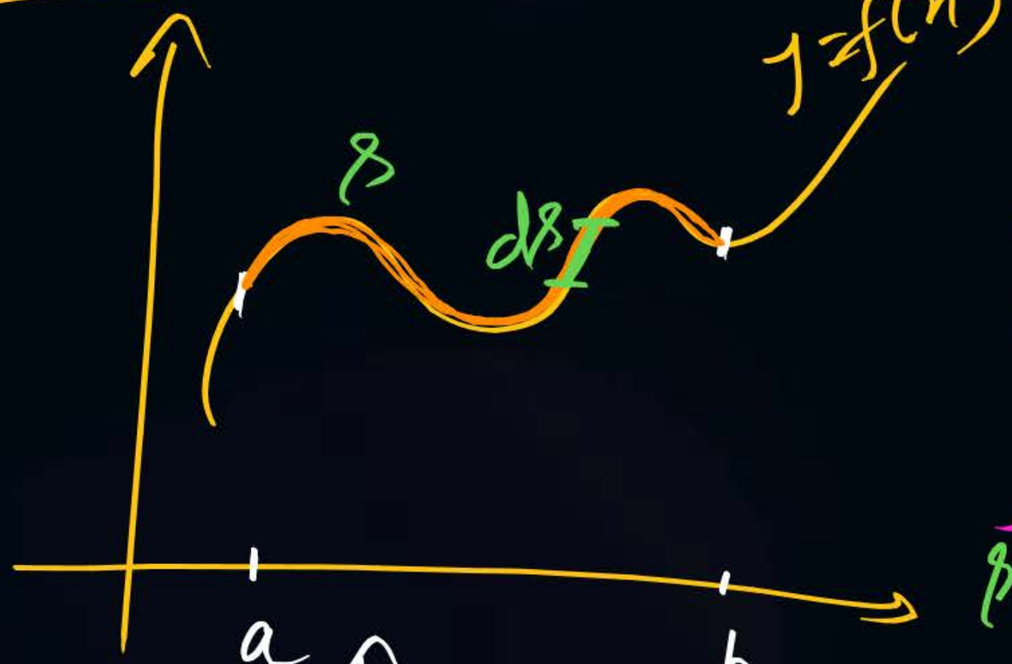
Length of Curve :-

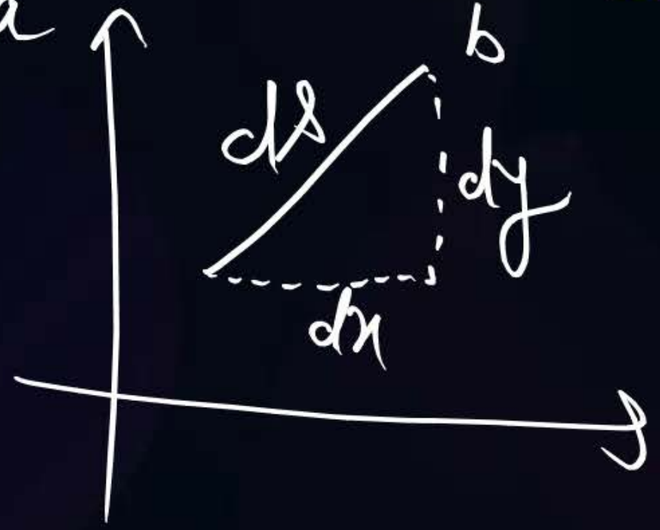
Case I: Length of Curve $y=f(x)$ b/n $x=a$ to b is $s=\int_{x=a}^{b}\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

Case II: " " $x=f(y)$ b/n $y=c$ to d is $s=\int_{y=c}^{d}\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$

Case II: If $x=x(t)$, $y=y(t)$, $z=z(t)$ then length of Curve b/n t_1 & t_2 is $s=\int_{t_1}^{t_2}\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt$

Proof (1)





$$(ds)^2 = (dx)^2 + (dy)^2$$

$$(ds)^2 = \left[1 + \left(\frac{dy}{dx}\right)^2\right](dx)^2$$

$$\int_{s=0}^{s} 1.\,ds = \int_{x=a}^{b} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,.\,dx$$

$$s = \int_{x=a}^{b} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,.\,dx$$

Q: Find the length of curve $y = \frac{2}{3}x^{3/2}$ b/n $x = 1$ & 2

Sol: $\frac{dy}{dx} = \frac{2}{3} \cdot \frac{3}{2} x^{\frac{3}{2}-1} = \sqrt{x}$

So Req length of curve,

$$s = \int_{x=1}^{2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$= \int_{x=1}^{2} \sqrt{1+x}\, dx$$

$$s = \left[\frac{(1+x)^{3/2}}{3/2}\right]_{x=1}^{2} = \frac{2}{3}\left[3^{3/2} - 2^{3/2}\right]$$

$$= \frac{2}{3}\left[\sqrt{27} - \sqrt{8}\right]$$ Ans.

(ii) Length of above curve b/n $x = 0$ to 1 ?

$$s = \frac{2}{3}\left[(1+x)^{3/2}\right]_0^1 = \frac{2}{3}\left[2^{3/2} - 1^{3/2}\right]$$

$$= \frac{2}{3}\left(\sqrt{8} - 1\right) = \frac{2}{3}\left(2\sqrt{2} - 1\right)$$ //

Q: If $x=\cos t$, $y=\sin t$, $z=\frac{2t}{\pi}$ then length of curve b/n $t=0$ & $t=\frac{\pi}{2}$

Sol: Req length $\beta=\int_{t=0}^{\pi/2}\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\cdot dt$

$$\beta=\int_0^{\pi/2}\sqrt{\sin^2 t+\cos^2 t+\frac{4}{\pi^2}}\cdot dt$$

$$=\sqrt{1+\frac{4}{\pi^2}}\int_0^{\pi/2}(1)\,dt=\frac{\sqrt{\pi^2+4}}{\pi}\left[\frac{\pi}{2}\right]=\frac{\sqrt{\pi^2+4}}{2}$$

Volume of solid formed by Revolution :-

Case I: (Revolution about y axis)

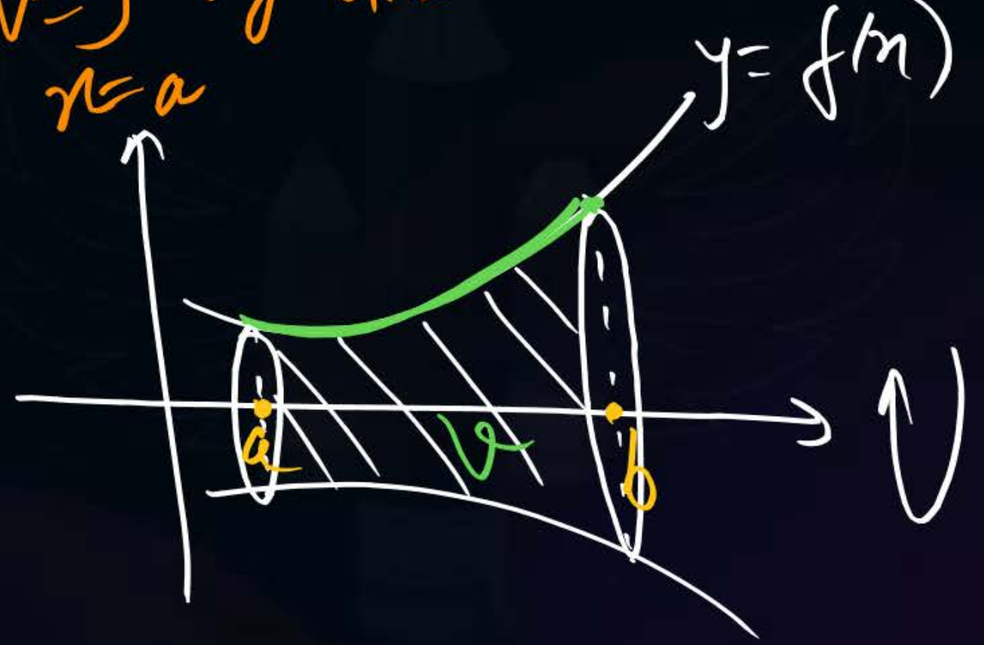
if the curve $x = f(y)$ is to be Revolved about y axis b/n $y = c$ to d then Volume of solid formed is

$$V = \int_{y=c}^{d} \pi x^2 \, dy$$

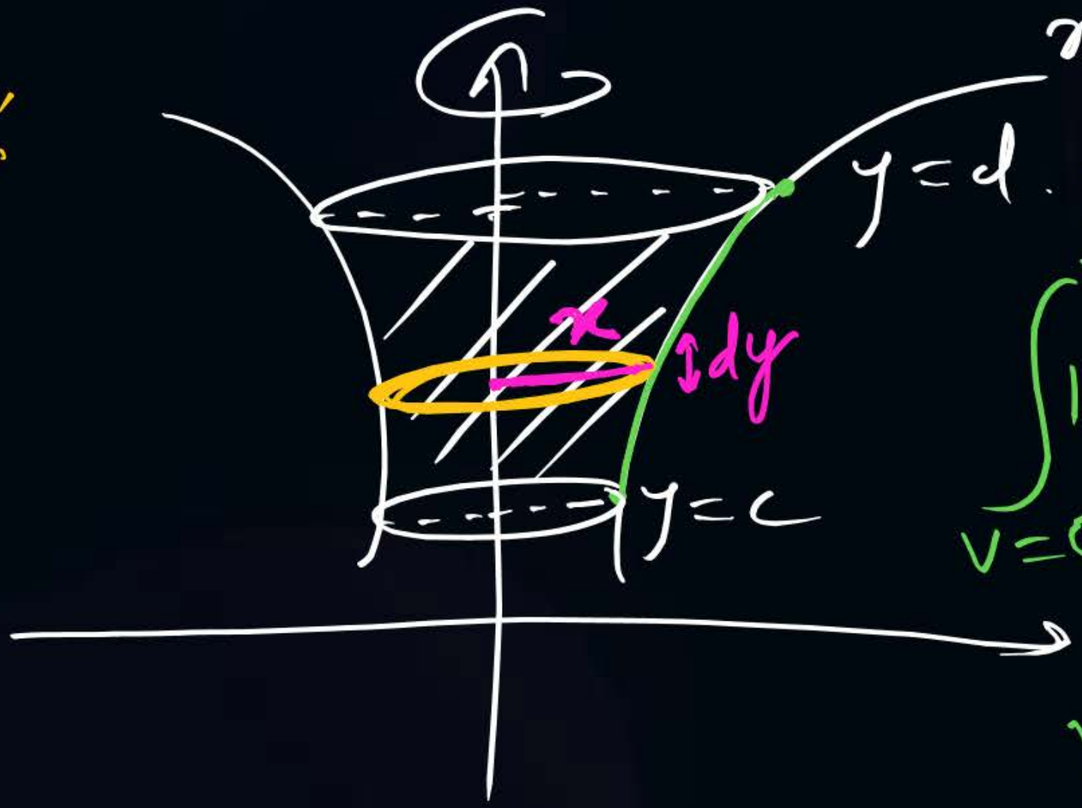
Case II: Revolution about x axis :-

if the curve $y = f(x)$ is to be Revolved about x axis b/n $x = a$ to b then Volume of solid formed is

$$V = \int_{x=a}^{b} \pi y^2 \, dx$$



Proof:



Volume of this small cylinder

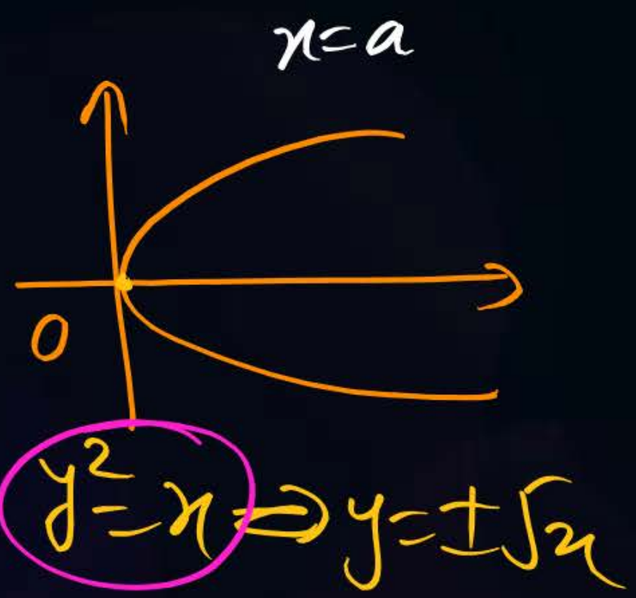
$$\int_{V=0}^{V} 1 \cdot dV = \int_{y=c}^{d} \pi (x^2)\, dy$$

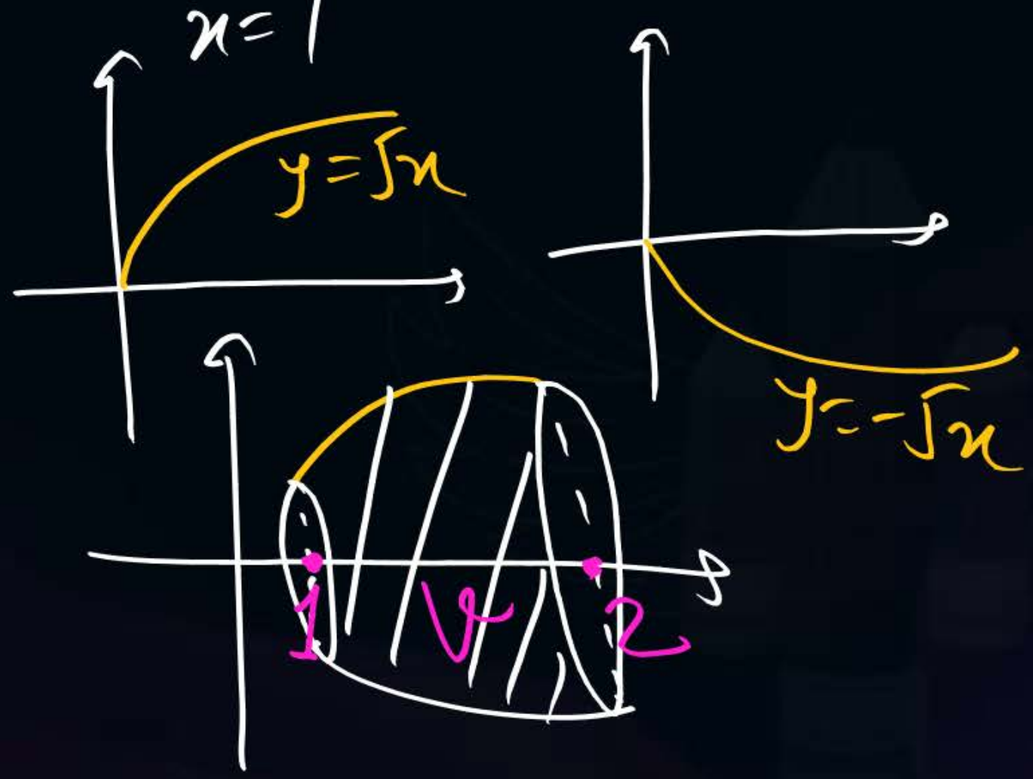
$$V = \int_{y=c}^{d} \pi x^2 dy$$

Q: Find the volume of solid formed by Revolution $y=\sqrt{x}$ b/n 1 & 2 about x axis ?

Sol: $\text{Req Volume} = \int_{x=a}^{b} \pi y^2 dx = \int_{1}^{2} \pi(x)\,dx = \pi\left(\frac{x^2}{2}\right)_1^2 = \frac{3\pi}{2}$

Note:





Q. $I = \int_0^{\pi/2} \log \sin x \, dx = ?$ ——— (1)

$= \int_0^{\pi/2} \log \sin\left(\frac{\pi}{2} - x\right) dx$ (By Prop 5)

$I = \int_0^{\pi/2} \log \cos x \, dx$ ——— (2)

By (1) & (2), $2I = \int_0^{\pi/2} (\log \sin x + \log \cos x) \, dx$

$2I = \int_0^{\pi/2} \log(\sin x \cos x) \, dx$

$= \int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right) dx$

$2I = \int_0^{\pi/2} \log \sin 2x \, dx - \int_0^{\pi/2} \log 2 \, dx$

$2I = I_1 - \frac{\pi}{2} \log 2$ ——— (3)

$I_1 = \int_0^{\pi/2} \log \sin 2x \, dx$

Put $2x = t$ ⟨ $t = 0$, $t = \pi$

$dx = \frac{dt}{2}$

$I_1 = \int_0^{\pi} \log \sin t \cdot \frac{dt}{2}$

$$I_1 = \int_0^{\pi} \log \sin t \, \frac{dt}{2} = \frac{2}{2}\int_0^{\pi/2} \log(\sin t)\, dt \quad \text{By Prop (6)}$$

$$= \int_0^{\pi/2} \log \sin t \, dt = \int_0^{\pi/2} \log \sin x \, dx = I \quad \text{By Prop (1)}$$

By (3) $\quad 2I = I_1 - \frac{\pi}{2}\log 2$

$$2I = I - \frac{\pi}{2}\log 2$$

$$I = -\frac{\pi}{2}\log 2 \quad \underline{\text{Ans}}$$

LINEAR ALGEBRA

CS/IT → 2 to 3.
DA → 14-16.

Linear Algebra – 1 (for Both CS/IT as well as for DA)

Linear Algebra – 2 (only for DA)

Determinants } 100% Q
Matrices
✓ RANK
S.O Eq.
L.D & L.I vectors

Evalues & E. Vectors
C.H.T
Diagonalisation
LU Decomposition

DETERMINANTS

Method of solving 2×2 Det :-

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \boxed{(a)(d) - (b)(c)}$$

eg $|A| = \begin{vmatrix} 2 & -3 \\ 1 & -4 \end{vmatrix} = ? = (2)(-4) - (1)(-3) = -5$

eg $|A| = \begin{vmatrix} (3+2i) & -i \\ i & (3-2i) \end{vmatrix} = ? = (3+2i)(3-2i) - (i)(-i)$

$= (3)^2 - (2i)^2 + i^2$

$= 9 - 4i^2 + i^2$

$= 9 - 3i^2 = 9 - 3(-1) = 12$

$i = \sqrt{-1}$ & $i^2 = -1$

Det of order 3×3 →

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

eg $|A| = \begin{vmatrix} 2 & 1 & -3 \\ 4 & 2 & 0 \\ -1 & 1 & 4 \end{vmatrix} = ?$

$= +(2)\begin{vmatrix} 2 & 0 \\ 1 & 4 \end{vmatrix} - (1)\begin{vmatrix} 4 & 0 \\ -1 & 4 \end{vmatrix} + (-3)\begin{vmatrix} 4 & 2 \\ -1 & 1 \end{vmatrix}$

Expanding along R_1

$= 2[8-0] - [16+0] - 3[4+2]$

$= -18$

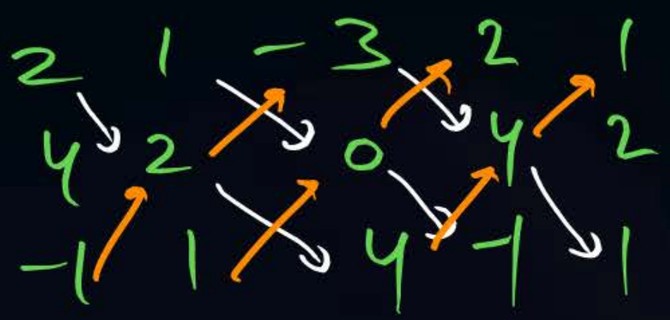
Expanding along R_2

$|A| = -(4)\begin{vmatrix} 1 & -3 \\ 1 & 4 \end{vmatrix} + (2)\begin{vmatrix} 2 & -3 \\ -1 & 4 \end{vmatrix} - (0)\begin{vmatrix} ? & ? \end{vmatrix}$

$= -4[4+3] + 2[8-3] - 0 = -18$

Shortcut of solving 3×3 Det (SARRUS Method) →

eg $|A| = \begin{vmatrix} 2 & 1 & -3 \\ 4 & 2 & 0 \\ -1 & 1 & 4 \end{vmatrix} = ?$



$|A| = (16 + 0 + (-12)) - (6 + 0 + 16)$

$= -18$

eg $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & -2 \\ 1 & -4 & 1 \end{vmatrix} = ?$



$|A| = (1 - 4 - 24) - (3 + 8 + 4)$

$= -42$

Method of Solving 4×4 Det →

$$\begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$$

eg $|A| = \begin{vmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{vmatrix} = ?$

2015 / 2M

Expanding along C_2

$$= -(1)\begin{vmatrix} 1 & 3 & 0 \\ 2 & 0 & 1 \\ 3 & 1 & 2 \end{vmatrix} + (0)\begin{vmatrix} = & = & = \\ = & = & = \\ = & = & = \end{vmatrix} - (3)\begin{vmatrix} 0 & 2 & 3 \\ 1 & 3 & 0 \\ 3 & 1 & 2 \end{vmatrix} + (0)\begin{vmatrix} = & = & = \\ = & = & = \\ = & = & = \end{vmatrix}$$

$$= -\left[-2\{6-0\} - 1\{1-9\}\right] + 0 - 3\left[-2\{2-0\} + 3\{1-9\}\right] + 0$$

$$= -\left[-12+8\right] - 3\left[-4-24\right] = 88 \quad \text{Ans}$$

Q. 2014 2M

$$|A| = \begin{vmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{vmatrix} = ?$$

(M-I) Solve by Conventional Approach → Do yourself. Ans = 5

(M-II) $R_1 \to R_1 - 2R_2$, $R_3 \to R_3 - R_2$, $R_4 \to R_4 - R_2$

$$|A| = \begin{vmatrix} 0 & -3 & -1 & -1 \\ 1 & 2 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{vmatrix}$$

Now Expanding along C_1,

$$|A| = -(1)\begin{vmatrix} -3 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = \cdots\cdots = 5$$

M-III (WRONG APPROACH) →

$$|A| = \begin{vmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{vmatrix} \xrightarrow[R_3 \to 2R_3 - R_1]{R_2 \to 2R_2 - R_1,\ R_4 \to 2R_4 - R_1} \begin{vmatrix} 2 & 1 & 1 & 1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 1 & 3 \end{vmatrix}$$

$$|A| = +(2)\begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{vmatrix} - 0 + 0 - 0 = \cdots = 40$$

$$|A| = 40 = 8 \times 5 = 2^3 \cdot (5)$$

Elementary Operation →

E-Row Operations

① $R_i \longleftrightarrow R_j$

② $R_i \longrightarrow kR_i$

③ $R_i \longrightarrow R_i + kR_j$

E-Column Operations

① $C_i \longleftrightarrow C_j$

② $C_i \longrightarrow kC_i$

③ $C_i \longrightarrow C_i + kC_j$

Note ① k = Any Scalar / Any Constant

② $R_3 \longleftrightarrow R_4$ ✓, $R_2 \to 5R_2$ ✓, $R_2 \to R_2 - 4R_3$ ✓, $R_3 \to 5R_3 - R_2$ ✗ it is mixed operation

(3) If we interchange any two Rows in a Matrix then value of it's Det changes by −ve sign.

(4) If we multiply any Row of a Matrix by K then Value of it's Det increases by K times.

(5) If we use 3rd E-operation in a Mat, then Value of it's Det remains unaltered. i.e

we are free to apply 3rd E-operation while calculating Det.

But take care Don't use Mixed operation while Calculating Det.

Sp Note- Above Prop are also Valid for Columns.

PROPERTIES of Determinants →

① If all the elements in a Row are Zero then Det = 0

② if any two Rows in a Mat are identical then it's Det = 0

eg: $|A| = \begin{vmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix}_{5\times5} = ?$

$R_1 \to R_1 + (R_2 + R_3 + R_4 + R_5)$

$= \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} = 0$

eg $|A| = \begin{vmatrix} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 31 & 32 & 33 & 34 \\ 41 & 42 & 43 & 44 \end{vmatrix}$

$R_4 \to R_4 - R_3$

$R_3 \to R_3 - R_2$

$R_2 \to R_2 - R_1$

$= \begin{vmatrix} 11 & 12 & 13 & 14 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \end{vmatrix} = 0$

THANK - YOU

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